

Year	S&P Composite Price Index	S&P Composite Dividends	S&P Composite Earnings	CPI Level (Dec 31st)	1 Year Treasury Yield (Annualized)	5 Year Treasury Yield (Annualized)	S&P Composite Price Index (2024)
1922	8.90						
1923	8.83	0.53	0.98	17.30			161.09
1924	10.58	0.55	0.93	17.30			193.01
1925	12.65	0.60	1.25	17.90			223.04
1926	13.40	0.69	1.24	17.70			238.93
1927	17.53	0.77	1.11	17.30			319.80
1928	24.86	0.85	1.38	17.10			458.83
1929	21.71	0.97	1.61	17.20			398.36
1930	15.98	0.98	0.97	16.10			313.25
1931	8.30	0.82	0.61	14.60			179.42
1932	7.09	0.50	0.41	13.10			170.81
1933	10.54	0.44	0.44	13.20	0.35%	1.35%	252.01
1934	9.26	0.45	0.49	13.40	0.28%	1.48%	218.10
1935	13.76	0.47	0.76	13.80	0.17%	1.35%	314.69
1936	17.59	0.72	1.02	14.00	0.17%	1.47%	396.54
1937	11.31	0.80	1.13	14.40	0.28%	1.58%	247.88
1938	12.50	0.51	0.64	14.00	0.07%	1.27%	281.79
1939	12.30	0.62	0.90	14.00	0.05%	1.35%	277.28
1940	10.55	0.67	1.05	14.10	0.04%	1.34%	236.14
1941	8.93	0.71	1.16	15.50	0.11%	1.48%	181.83
1942	10.09	0.59	1.03	16.90	0.43%	1.73%	188.43
1943	11.85	0.61	0.94	17.40	0.84%	1.65%	214.94
1944	13.49	0.64	0.93	17.80	0.87%	1.50%	239.19
1945	18.02	0.66	0.96	18.20	0.79%	1.27%	312.48
1946	15.21	0.71	1.06	21.50	0.83%	0.99%	223.27
1947	14.83	0.84	1.61	23.40	0.96%	1.16%	200.02
1948	15.36	0.93	2.29	24.10	1.07%	1.44%	201.15
1949	16.88	1.14	2.32	23.60	1.28%	1.57%	225.74
1950	21.21	1.47	2.84	25.00	1.05%	1.26%	267.76
1951	24.19	1.41	2.44	26.50	1.47%	1.50%	288.09
1952	26.18	1.41	2.40	26.70	1.78%	2.16%	309.46
1953	25.46	1.45	2.51	26.90	1.95%	2.36%	298.71
1954	35.60	1.54	2.77	26.70	1.54%	2.19%	420.81
1955	44.15	1.64	3.62	26.80	1.21%	2.16%	519.92
1956	45.43	1.74	3.41	27.60	2.73%	2.93%	519.49
1957	41.12	1.79	3.37	28.40	3.68%	3.70%	456.96
1958	55.62	1.75	2.89	28.90	3.18%	3.08%	607.40
1959	58.03	1.83	3.39	29.40	3.29%	3.82%	622.94
1960	59.72	1.95	3.27	29.80	5.14%	5.01%	632.48
1961	69.07	2.02	3.19	30.00	2.86%	3.67%	726.63
1962	65.06	2.13	3.67	30.40	3.18%	3.91%	675.44
1963	76.45	2.28	4.02	30.90	3.01%	3.56%	780.84

1964	86.12	2.50	4.55	31.20	3.81%	4.04%	871.15
1965	93.32	2.72	5.19	31.80	4.02%	4.09%	926.17
1966	84.45	2.87	5.55	32.90	4.72%	4.72%	810.12
1967	95.04	2.92	5.33	33.90	5.20%	5.00%	884.81
1968	102.04	3.07	5.76	35.50	5.71%	5.75%	907.16
1969	90.31	3.16	5.78	37.70	6.19%	6.12%	756.03
1970	93.49	3.14	5.13	39.80	8.17%	7.96%	741.35
1971	103.30	3.07	5.70	41.10	5.00%	5.95%	793.24
1972	118.42	3.15	6.42	42.50	4.60%	5.69%	879.39
1973	96.11	3.38	8.16	46.20	5.52%	6.16%	656.55
1974	72.56	3.60	8.89	51.90	7.27%	6.80%	441.24
1975	96.86	3.68	7.96	55.50	7.31%	7.31%	550.80
1976	103.81	4.05	9.91	58.20	6.60%	7.76%	562.94
1977	90.25	4.67	10.89	62.10	4.89%	6.10%	458.67
1978	99.71	5.07	12.33	67.70	6.96%	7.48%	464.83
1979	110.87	5.65	14.86	76.70	10.30%	9.08%	456.21
1980	132.97	6.16	14.82	86.30	11.98%	10.42%	486.28
1981	117.28	6.63	15.36	94.00	14.88%	13.25%	393.77
1982	144.27	6.87	12.64	97.60	12.85%	13.60%	466.52
1983	166.39	7.09	14.03	101.30	8.91%	10.22%	518.40
1984	171.61	7.53	16.64	105.30	10.11%	11.54%	514.35
1985	208.19	7.90	14.61	109.30	9.33%	11.07%	601.15
1986	264.51	8.28	14.48	110.50	7.67%	8.73%	755.48
1987	250.48	8.81	17.50	115.40	5.87%	6.67%	685.03
1988	285.41	9.73	23.76	120.50	7.17%	8.45%	747.53
1989	339.97	11.05	22.87	126.10	8.99%	9.09%	850.88
1990	325.50	12.10	21.34	133.80	7.72%	7.75%	767.78
1991	416.08	12.20	15.91	137.90	7.05%	7.73%	952.26
1992	435.23	12.38	19.09	141.90	4.38%	6.19%	968.01
1993	472.99	12.58	21.88	145.80	3.71%	6.08%	1023.85
1994	465.25	13.18	30.60	149.70	3.61%	5.15%	980.86
1995	614.42	13.79	33.96	153.50	7.14%	7.78%	1263.28
1996	766.22	14.90	38.73	158.60	5.31%	5.51%	1524.73
1997	963.36	15.50	39.72	161.30	5.47%	6.07%	1884.94
1998	1248.77	16.20	37.70	163.90	5.53%	5.77%	2404.63
1999	1425.59	16.69	48.17	168.30	4.52%	4.45%	2673.34
2000	1330.93	16.27	50.00	174.00	5.84%	6.19%	2414.07
2001	1140.21	15.74	24.69	176.70	5.60%	5.17%	2036.54
2002	895.84	16.07	27.59	180.90	2.22%	4.39%	1562.92
2003	1080.64	17.39	48.74	184.30	1.45%	3.03%	1850.54
2004	1199.21	19.44	58.55	190.30	1.31%	3.27%	1988.84
2005	1262.07	22.22	69.93	196.80	2.67%	3.60%	2023.96
2006	1416.42	24.88	81.51	201.80	4.35%	4.39%	2215.21
2007	1479.22	27.73	66.18	210.04	4.94%	4.53%	2222.71

2008	877.56	28.39	26.16	210.23	3.26%	3.49%	1317.44
2009	1110.38	22.41	50.97	215.95	0.49%	1.52%	1622.80
2010	1241.53	22.73	77.35	219.18	0.37%	2.34%	1787.73
2011	1243.32	26.43	86.95	225.67	0.29%	1.93%	1738.80
2012	1422.29	31.25	86.51	229.60	0.12%	0.89%	1955.05
2013	1807.78	34.99	100.20	233.05	0.16%	0.70%	2448.17
2014	2054.27	39.44	102.31	234.81	0.13%	1.58%	2761.09
2015	2054.08	43.39	86.53	236.53	0.21%	1.64%	2740.84
2016	2246.63	45.70	94.55	241.43	0.65%	1.70%	2936.84
2017	2664.34	48.93	109.88	246.52	0.87%	1.96%	3410.94
2018	2567.31	53.75	132.39	251.23	1.70%	2.18%	3225.12
2019	3176.75	58.24	139.47	256.97	2.66%	2.68%	3901.55
2020	3695.31	58.28	94.13	260.47	1.55%	1.68%	4477.45
2021	4674.77	60.40	197.87	278.80	0.10%	0.36%	5291.86
2022	3912.38	66.92	172.75	296.80	0.40%	1.27%	4160.31
2023	4685.05	70.30	192.43	306.75	4.71%	3.99%	4820.36
2024	6010.91	74.83	210.17	315.61	4.79%	3.90%	6010.91
Average							

S&P Composite Dividends (2024)	S&P Composite Earnings (2024)	10-Year Avg Real Earnings	Real CAPE	Nominal Annual Return	Real Annual Return	Future 1-Year Nominal Return	Future 1-Year Real Return	Future 5- Year Nominal Return
9.67	17.88			5.17%		26.05%	26.05%	28.63%
10.03	16.97			26.05%	26.05%	25.24%	21.04%	20.58%
10.58	22.04			25.24%	21.04%	11.38%	12.64%	9.72%
12.30	22.11			11.38%	12.64%	36.57%	39.72%	-4.02%
14.05	20.25			36.57%	39.72%	46.66%	48.38%	-11.41%
15.69	25.47			46.66%	48.38%	-8.77%	-9.30%	-10.45%
17.80	29.54			-8.77%	-9.30%	-21.88%	-16.54%	-10.27%
19.21	19.01			-21.88%	-16.54%	-42.93%	-37.07%	2.73%
17.73	13.19			-42.93%	-37.07%	-8.55%	1.92%	21.68%
12.05	9.88	19.63	8.70	-8.55%	1.92%	54.87%	53.69%	14.97%
10.52	10.52	18.90	13.34	54.87%	53.69%	-7.87%	-9.25%	8.33%
10.60	11.54	18.36	11.88	-7.87%	-9.25%	53.67%	49.22%	10.85%
10.75	17.38	17.89	17.59	53.67%	49.22%	33.07%	31.17%	-0.13%
16.23	22.99	17.98	22.06	33.07%	31.17%	-31.15%	-33.07%	-7.36%
17.53	24.77	18.43	13.45	-31.15%	-33.07%	15.03%	18.32%	3.46%
11.50	14.43	17.33	16.26	15.03%	18.32%	3.36%	3.36%	4.94%
13.98	20.29	16.40	16.91	3.36%	3.36%	-8.78%	-9.43%	7.98%
15.00	23.50	16.85	14.02	-8.78%	-9.43%	-8.63%	-16.88%	17.38%
14.46	23.62	17.89	10.16	-8.63%	-16.88%	19.60%	9.69%	16.59%
11.02	19.24	18.83	10.01	19.60%	9.69%	23.49%	19.94%	13.16%
11.06	17.05	19.48	11.03	23.49%	19.94%	19.24%	16.56%	10.55%
11.35	16.49	19.98	11.97	19.24%	16.56%	38.47%	35.43%	10.19%
11.45	16.65	19.90	15.70	38.47%	35.43%	-11.65%	-25.21%	9.52%
10.42	15.56	19.16	11.65	-11.65%	-25.21%	3.02%	-5.34%	16.58%
11.33	21.71	18.85	10.61	3.02%	-5.34%	9.84%	6.65%	18.97%
12.18	29.99	20.41	9.86	9.84%	6.65%	17.32%	19.80%	17.40%
15.25	31.03	21.48	10.51	17.32%	19.80%	34.36%	26.84%	22.63%
18.56	35.85	22.72	11.79	34.36%	26.84%	20.70%	13.87%	21.57%
16.79	29.06	23.26	12.38	20.70%	13.87%	14.06%	13.20%	18.64%
16.67	28.37	24.18	12.80	14.06%	13.20%	2.79%	2.02%	14.25%
17.01	29.45	25.42	11.75	2.79%	2.02%	45.88%	46.97%	21.44%
18.20	32.74	27.04	15.56	45.88%	46.97%	28.62%	28.14%	14.28%
19.31	42.63	29.64	17.54	28.62%	28.14%	6.84%	3.74%	10.00%
19.90	38.99	31.98	16.24	6.84%	3.74%	-5.55%	-8.21%	12.40%
19.89	37.45	33.56	13.62	-5.55%	-8.21%	39.52%	37.10%	13.07%
19.11	31.56	33.71	18.02	39.52%	37.10%	7.62%	5.79%	9.89%
19.64	36.39	34.25	18.19	7.62%	5.79%	6.27%	4.85%	11.54%
20.65	34.63	34.13	18.53	6.27%	4.85%	19.04%	18.25%	12.62%
21.25	33.56	34.58	21.01	19.04%	18.25%	-2.72%	-4.00%	7.32%
22.11	38.10	35.55	19.00	-2.72%	-4.00%	21.01%	19.05%	11.17%
23.29	41.06	36.71	21.27	21.01%	19.05%	15.92%	14.80%	9.19%

25.29	46.03	38.04	22.90	15.92%	14.80%	11.52%	9.41%	4.16%
27.00	51.51	38.93	23.79	11.52%	9.41%	-6.43%	-9.56%	3.30%
27.53	53.24	40.35	20.08	-6.43%	-9.56%	16.00%	12.58%	7.42%
27.18	49.62	41.57	21.28	16.00%	12.58%	10.60%	5.61%	7.74%
27.29	51.21	43.53	20.84	10.60%	5.61%	-8.40%	-13.74%	1.97%
26.45	48.39	44.73	16.90	-8.40%	-13.74%	7.00%	1.35%	-0.94%
24.90	40.68	45.34	16.35	7.00%	1.35%	13.78%	10.18%	4.32%
23.57	43.77	46.36	17.11	13.78%	10.18%	17.69%	13.81%	3.87%
23.39	47.67	47.32	18.58	17.69%	13.81%	-15.99%	-22.71%	-1.25%
23.09	55.74	48.79	13.46	-15.99%	-22.71%	-20.76%	-29.46%	5.35%
21.89	54.06	49.59	8.90	-20.76%	-29.46%	38.56%	29.57%	13.87%
20.93	45.27	48.97	11.25	38.56%	29.57%	11.36%	6.19%	11.63%
21.96	53.74	49.01	11.49	11.36%	6.19%	-8.56%	-14.31%	7.72%
23.73	55.35	49.59	9.25	-8.56%	-14.31%	16.10%	6.50%	15.38%
23.64	57.48	50.21	9.26	16.10%	6.50%	16.86%	3.15%	16.19%
23.25	61.15	51.49	8.86	16.86%	3.15%	25.49%	11.53%	14.30%
22.53	54.20	52.84	9.20	25.49%	11.53%	-6.81%	-14.45%	14.38%
22.26	51.57	53.62	7.34	-6.81%	-14.45%	28.87%	24.12%	22.45%
22.22	40.87	52.94	8.81	28.87%	24.12%	20.25%	15.85%	15.93%
22.09	43.71	51.74	10.02	20.25%	15.85%	7.66%	3.57%	15.46%
22.57	49.87	51.32	10.02	7.66%	3.57%	25.92%	21.31%	18.57%
22.81	42.19	51.01	11.78	25.92%	21.31%	31.03%	29.61%	13.07%
23.65	41.36	49.77	15.18	31.03%	29.61%	-1.97%	-6.14%	13.17%
24.09	47.86	49.03	13.97	-1.97%	-6.14%	17.83%	12.84%	15.29%
25.48	62.23	49.50	15.10	17.83%	12.84%	22.99%	17.53%	14.04%
27.66	57.24	49.11	17.33	22.99%	17.53%	-0.70%	-6.41%	9.67%
28.54	50.34	48.72	15.76	-0.70%	-6.41%	31.58%	27.66%	16.62%
27.92	36.41	47.21	20.17	31.58%	27.66%	7.58%	4.55%	15.82%
27.53	42.46	47.37	20.44	7.58%	4.55%	11.57%	8.58%	19.87%
27.23	47.36	47.73	21.45	11.57%	8.58%	1.15%	-1.49%	23.84%
27.79	64.51	49.20	19.94	1.15%	-1.49%	35.03%	31.68%	27.17%
28.35	69.82	51.96	24.31	35.03%	31.68%	27.13%	23.04%	18.41%
29.65	77.07	55.53	27.46	27.13%	23.04%	27.75%	25.61%	9.72%
30.33	77.72	58.52	32.21	27.75%	25.61%	31.31%	29.23%	-0.09%
31.19	72.59	59.55	40.38	31.31%	29.23%	15.50%	12.48%	-1.46%
31.30	90.33	62.86	42.53	15.50%	12.48%	-5.50%	-8.59%	-1.93%
29.51	90.69	66.90	36.09	-5.50%	-8.59%	-13.15%	-14.47%	0.56%
28.11	44.10	67.67	30.10	-13.15%	-14.47%	-20.02%	-21.88%	6.22%
28.04	48.13	68.23	22.91	-20.02%	-21.88%	22.57%	20.31%	12.46%
29.78	83.46	71.84	25.76	22.57%	20.31%	12.77%	9.22%	-2.11%
32.24	97.10	75.10	26.48	12.77%	9.22%	7.09%	3.56%	0.57%
35.63	112.15	79.34	25.51	7.09%	3.56%	14.20%	11.37%	1.81%
38.91	127.48	84.38	26.25	14.20%	11.37%	6.39%	2.22%	-0.41%
41.67	99.44	86.55	25.68	6.39%	2.22%	-38.75%	-38.81%	1.48%

42.62	39.27	83.22	15.83	-38.75%	-38.81%	29.08%	25.66%	17.89%
32.75	74.49	81.63	19.88	29.08%	25.66%	13.86%	12.18%	15.36%
32.73	111.38	83.70	21.36	13.86%	12.18%	2.27%	-0.67%	12.87%
36.96	121.60	91.45	19.01	2.27%	-0.67%	16.91%	14.91%	14.86%
42.96	118.91	98.53	19.84	16.91%	14.91%	29.56%	27.65%	15.61%
47.38	135.70	103.75	23.60	29.56%	27.65%	15.82%	14.95%	9.41%
53.01	137.51	107.79	25.61	15.82%	14.95%	2.10%	1.36%	11.27%
57.90	115.46	108.12	25.35	2.10%	1.36%	11.60%	9.33%	14.57%
59.74	123.60	107.74	27.26	11.60%	9.33%	20.77%	18.28%	17.78%
62.64	140.67	111.86	30.49	20.77%	18.28%	-1.62%	-3.47%	9.82%
67.52	166.31	124.56	25.89	-1.62%	-3.47%	26.01%	23.19%	14.57%
71.53	171.29	134.24	29.06	26.01%	23.19%	18.16%	16.57%	15.27%
70.61	114.05	134.51	33.29	18.16%	16.57%	28.14%	19.72%	
68.37	223.99	144.75	36.56	28.14%	19.72%	-14.88%	-20.04%	
71.16	183.70	151.23	27.51	-14.88%	-20.04%	21.55%	17.60%	
72.33	197.99	157.46	30.61	21.55%	17.60%	29.90%	26.25%	
74.83	210.17	164.72	36.49	29.90%	26.25%			
			18.96	12.15%	9.10%			

**Future
5-Year
Real
Return**

Previous Year CAPE

S&P 500 Weight

T-bill Weight

28.93%

20.72%

12.07%

-0.25%

-6.35%

-5.69%

-5.68%

5.95%

22.71%

12.81%

7.06%

8.70

0.8

0.2

9.88%

13.34

0.8

0.2

-0.56%

11.88

0.8

0.2

-9.23%

17.59

0.6

0.4

0.20%

22.06

0.6

0.4

0.47%

13.45

0.8

0.2

2.91%

16.26

0.6

0.4

11.54%

16.91

0.6

0.4

9.21%

14.02

0.8

0.2

6.03%

10.16

0.8

0.2

3.57%

10.01

0.8

0.2

4.14%

11.03

0.8

0.2

2.79%

11.97

0.8

0.2

11.80%

15.70

0.6

0.4

15.87%

11.65

0.8

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14.85%

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19.64%

9.86

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19.89%

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17.68%

11.79

0.8

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12.84%

12.38

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19.71%

12.80

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12.10%

11.75

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7.69%

15.56

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17.54

0.6

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11.54%

16.24

0.6

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8.43%

13.62

0.8

0.2

10.22%

18.02

0.6

0.4

11.16%

18.19

0.6

0.4

5.36%

18.53

0.6

0.4

8.77%

21.01

0.6

0.4

6.20%

19.00

0.6

0.4

0.29%	21.27	0.6	0.4
-1.23%	22.90	0.6	0.4
2.75%	23.79	0.6	0.4
2.97%	20.08	0.6	0.4
-3.26%	21.28	0.6	0.4
-7.08%	20.84	0.6	0.4
-2.40%	16.90	0.6	0.4
-3.11%	16.35	0.6	0.4
-8.46%	17.11	0.6	0.4
-2.40%	18.58	0.6	0.4
5.31%	13.46	0.8	0.2
2.20%	8.90	0.8	0.2
-2.12%	11.25	0.8	0.2
5.40%	11.49	0.8	0.2
7.19%	9.25	0.8	0.2
7.28%	9.26	0.8	0.2
9.10%	8.86	0.8	0.2
18.55%	9.20	0.8	0.2
12.11%	7.34	0.8	0.2
11.52%	8.81	0.8	0.2
14.37%	10.02	0.8	0.2
8.59%	10.02	0.8	0.2
8.26%	11.78	0.8	0.2
10.62%	15.18	0.6	0.4
9.77%	13.97	0.8	0.2
5.97%	15.10	0.6	0.4
13.46%	17.33	0.6	0.4
12.62%	15.76	0.6	0.4
16.84%	20.17	0.6	0.4
20.97%	20.44	0.6	0.4
24.22%	21.45	0.6	0.4
15.48%	19.94	0.6	0.4
7.37%	24.31	0.6	0.4
-2.36%	27.46	0.3	0.7
-3.74%	32.21	0.3	0.7
-4.31%	40.38	0.3	0.7
-1.89%	42.53	0.3	0.7
3.43%	36.09	0.3	0.7
9.15%	30.10	0.3	0.7
-4.66%	22.91	0.6	0.4
-1.94%	25.76	0.3	0.7
-0.36%	26.48	0.3	0.7
-2.62%	25.51	0.3	0.7
-0.31%	26.25	0.3	0.7

15.48%	25.68	0.3	0.7
13.44%	15.83	0.6	0.4
11.16%	19.88	0.6	0.4
13.32%	21.36	0.6	0.4
13.97%	19.01	0.6	0.4
7.78%	19.84	0.6	0.4
9.28%	23.60	0.6	0.4
12.38%	25.61	0.3	0.7
14.44%	25.35	0.3	0.7
5.82%	27.26	0.3	0.7
10.08%	30.49	0.3	0.7
10.63%	25.89	0.3	0.7
	29.06	0.3	0.7
	33.29	0.3	0.7
	36.56	0.3	0.7
	27.51	0.3	0.7
	30.61	0.3	0.7
	18.77	0.60	0.40

Based on the regression results, CAPE has a negative relationship with future return. I use the previous year's CAPE ratio as a signal for asset allocation. When CAPE is above 25, I allocate 20% to Treasury bills. When CAPE is between 15 and 25, I allocate 60% to equity and 40% to Treasury bills. When CAPE is below 15, I allocate 100% to Treasury bills. When CAPE is above 25, indicating overvaluation, I reduce equity exposure to 30% and increase Treasury bill exposure to 70%. This strategy is based on the idea that investors should increase equity exposure when the market is undervalued and reduce equity exposure when the market is overvalued. Based on my calculations, the strategy generates an average return of 10.5% per year. Therefore, the CAPE-based market-timing strategy is a viable investment strategy.

Strategy Return 60/40 Benchmark Return

43.96%	33%
-6.24%	-5%
42.97%	32%
19.91%	20%
-18.58%	-19%
12.04%	9%
2.03%	2%
-5.25%	-5%
-6.88%	-5%
15.76%	12%
18.96%	14%
15.57%	12%
30.94%	23%
-6.66%	-7%
2.61%	2%
8.09%	6%
14.11%	11%
27.70%	21%
16.85%	13%
11.60%	9%
2.62%	2%
37.01%	28%
17.66%	18%
5.20%	5%
-1.86%	-2%
32.25%	25%
5.89%	6%
5.82%	6%
12.57%	13%
-0.36%	0%
13.81%	14%

11.08%	11%
8.52%	9%
-1.97%	-2%
11.68%	12%
8.64%	9%
-2.56%	-3%
7.47%	7%
10.27%	10%
12.45%	12%
-7.38%	-7%
-15.15%	-10%
32.31%	26%
10.41%	9%
-5.87%	-3%
14.27%	12%
15.55%	14%
22.79%	20%
-2.47%	2%
25.67%	22%
17.98%	16%
8.15%	9%
22.60%	19%
26.36%	22%
1.16%	1%
15.70%	14%
17.39%	17%
2.67%	3%
21.77%	22%
6.30%	6%
8.42%	8%
2.13%	2%
23.87%	24%
18.40%	18%
12.15%	19%
13.26%	21%
7.81%	11%
2.44%	-1%
-0.02%	-6%
-4.45%	-11%
14.12%	14%
4.75%	8%
4.00%	5%
7.31%	10%
5.38%	6%

-9.34%	-22%
17.65%	18%
8.46%	8%
1.48%	1%
10.19%	10%
17.80%	18%
9.54%	10%
0.78%	1%
3.93%	7%
6.84%	13%
0.70%	0%
9.66%	17%
6.53%	12%
8.51%	17%
-4.18%	-9%
9.76%	15%
12.32%	20%
10%	9%

ns and stronger predictive power over longer horizons. Therefore, I
s low, indicating undervaluation, I allocate 80% to the S&P 500 and
ities and 40% to Treasury bills. When CAPE is high, indicating
and invest 70% in Treasury bills.

when the market is undervalued and reduce exposure when it is
about 4%, compared to about 3% for the static 60/40 benchmark.
slightly outperforms the benchmark.

Y1real

Summary Output

Regression Statistics	
Multiple R	0.181589393
R Square	0.032974708
Adjusted R Square	0.022109255
Standard Error	0.170603058
Observations	91

ANOVA

	df	SS	MS	F	Significance F
Regression	1	0.088329699	0.088329699	3.034821342	0.084950167
Residual	89	2.590380892	0.029105403		
total	90	2.678710591			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	0.162322846	0.04752849	3.415274666	0.000961728	0.06788475	0.256760942
8.700066785	-0.004062539	0.002332013	-1.742073862	0.084950167	-0.0086962	0.000571121

Y1normal

Summary Output

Regression Statistics	
Multiple R	0.235878417
R Square	0.055638627
Adjusted R Square	0.045027826
Standard Error	0.165360098
Observations	91

ANOVA

	df	SS	MS	F	Significance F
Regression	1	0.143380353	0.143380353	5.24358364	0.024392282
Residual	89	2.433612633	0.027343962		
total	90	2.576992986			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	0.220337418	0.046067848	4.782889331	6.80487E-06	0.128801586	0.31187325
8.700066785	-0.005175938	0.002260346	-2.289887255	0.024392282	-0.009667197	-0.000684678

The regression results indicate a negative relationship between CAPE and future stock returns, as evidenced by the negative coefficient. However, the predictive power is weak for 1-year returns, as the coefficient is not statistically significant, suggesting that the relationship is likely driven by market noise.

In contrast, CAPE shows stronger predictive power for 5-year returns, with a more statistically significant relationship. This suggests that market effects take time to materialize, as mean reversion dominates in the long run. Economically, a high CAPE ratio reflects overvaluation associated with lower future returns, while a low CAPE indicates undervaluation and higher expected future returns.

Y5real

Summary Output

Regression Statistics	
Multiple R	0.358449233
R Square	0.128485853
Adjusted R Square	0.118232745
Standard Error	0.069021464
Observations	87

ANOVA

	df	SS
Regression	1	0.059699147
Residual	85	0.404936808
total	86	0.464635954

Lower 95.0%	Upper 95.0%
0.06788475	0.256760942
-0.0086962	0.000571121

		Coefficients	Standard Error
Intercept		0.137627497	0.020042775
	8.700066785	-0.003607062	0.001018952

y5 normal

Summary Output

Regression Statistics	
Multiple R	0.502068811
R Square	0.252073091
Adjusted R Square	0.243273951
Standard Error	0.060943146
Observations	87

ANOVA

	df	SS
Regression	1	0.10639862
Residual	85	0.315695699
total	86	0.422094318

Lower 95.0%	Upper 95.0%
0.128801586	0.31187325
-0.009667197	-0.000684678

		Coefficients	Standard Error
Intercept		0.197598804	0.017696955
	8.700066785	-0.004815457	0.000899693

egative coefficient on CAPE.
at short-term returns are largely

). This suggests that valuation
an overvalued market, which is
ed returns.

MS	F	Significance F
0.059699147	12.53140586	0.000651868
0.004763962		

t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Jpper 95.0%
6.866688765	1.0137E-09	0.097777091	0.177477903	0.097777091	0.1774779
-3.539972578	0.000651868	-0.005633011	-0.001581112	-0.005633011	-0.00158111

MS	F	Significance F
0.10639862	28.64746877	7.25204E-07
0.003714067		

t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Jpper 95.0%
11.16569496	2.40031E-18	0.162412516	0.232785092	0.162412516	0.23278509
-5.352333021	7.25204E-07	-0.006604288	-0.003026626	-0.006604288	-0.00302663

Previous Year CAPE	S&P 500 Weight	T-bill Weight	Strategy Return
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8.70	0.8	0.2	43.96%
13.34	0.8	0.2	-6.24%
11.88	0.8	0.2	42.97%
17.59	0.6	0.4	19.91%
22.06	0.6	0.4	-18.58%
13.45	0.8	0.2	12.04%
16.26	0.6	0.4	2.03%
16.91	0.6	0.4	-5.25%
14.02	0.8	0.2	-6.88%
10.16	0.8	0.2	15.76%
10.01	0.8	0.2	18.96%
11.03	0.8	0.2	15.57%
11.97	0.8	0.2	30.94%
15.70	0.6	0.4	-6.66%
11.65	0.8	0.2	2.61%
10.61	0.8	0.2	8.09%
9.86	0.8	0.2	14.11%
10.51	0.8	0.2	27.70%
11.79	0.8	0.2	16.85%
12.38	0.8	0.2	11.60%
12.80	0.8	0.2	2.62%
11.75	0.8	0.2	37.01%
15.56	0.6	0.4	17.66%
17.54	0.6	0.4	5.20%
16.24	0.6	0.4	-1.86%
13.62	0.8	0.2	32.25%
18.02	0.6	0.4	5.89%
18.19	0.6	0.4	5.82%
18.53	0.6	0.4	12.57%
21.01	0.6	0.4	-0.36%
19.00	0.6	0.4	13.81%

21.27	0.6	0.4	11.08%
22.90	0.6	0.4	8.52%
23.79	0.6	0.4	-1.97%
20.08	0.6	0.4	11.68%
21.28	0.6	0.4	8.64%
20.84	0.6	0.4	-2.56%
16.90	0.6	0.4	7.47%
16.35	0.6	0.4	10.27%
17.11	0.6	0.4	12.45%
18.58	0.6	0.4	-7.38%
13.46	0.8	0.2	-15.15%
8.90	0.8	0.2	32.31%
11.25	0.8	0.2	10.41%
11.49	0.8	0.2	-5.87%
9.25	0.8	0.2	14.27%
9.26	0.8	0.2	15.55%
8.86	0.8	0.2	22.79%
9.20	0.8	0.2	-2.47%
7.34	0.8	0.2	25.67%
8.81	0.8	0.2	17.98%
10.02	0.8	0.2	8.15%
10.02	0.8	0.2	22.60%
11.78	0.8	0.2	26.36%
15.18	0.6	0.4	1.16%
13.97	0.8	0.2	15.70%
15.10	0.6	0.4	17.39%
17.33	0.6	0.4	2.67%
15.76	0.6	0.4	21.77%
20.17	0.6	0.4	6.30%
20.44	0.6	0.4	8.42%
21.45	0.6	0.4	2.13%
19.94	0.6	0.4	23.87%
24.31	0.6	0.4	18.40%
27.46	0.3	0.7	12.15%
32.21	0.3	0.7	13.26%
40.38	0.3	0.7	7.81%
42.53	0.3	0.7	2.44%
36.09	0.3	0.7	-0.02%
30.10	0.3	0.7	-4.45%
22.91	0.6	0.4	14.12%
25.76	0.3	0.7	4.75%
26.48	0.3	0.7	4.00%
25.51	0.3	0.7	7.31%
26.25	0.3	0.7	5.38%

25.68	0.3	0.7	-9.34%
15.83	0.6	0.4	17.65%
19.88	0.6	0.4	8.46%
21.36	0.6	0.4	1.48%
19.01	0.6	0.4	10.19%
19.84	0.6	0.4	17.80%
23.60	0.6	0.4	9.54%
25.61	0.3	0.7	0.78%
25.35	0.3	0.7	3.93%
27.26	0.3	0.7	6.84%
30.49	0.3	0.7	0.70%
25.89	0.3	0.7	9.66%
29.06	0.3	0.7	6.53%
33.29	0.3	0.7	8.51%
36.56	0.3	0.7	-4.18%
27.51	0.3	0.7	9.76%
30.61	0.3	0.7	12.32%
18.77	0.60	0.40	10%

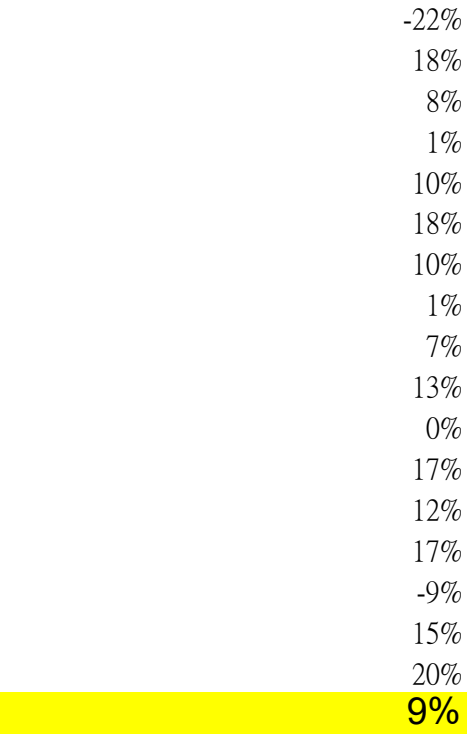
Based on the regression results, CAPE has a negative relationship with future returns and stronger predictive power than the previous year's CAPE ratio as a signal for asset allocation. When CAPE is low, indicating undervaluation, I allocate 80% to equities and 20% to Treasury bills. When CAPE is between 15 and 25, I allocate 60% to equities and 40% to Treasury bills. When CAPE is high, indicating overvaluation, I reduce equity exposure to 30% and invest 70% in Treasury bills.

This strategy is based on the idea that investors should increase equity exposure when the market is undervalued and decrease equity exposure when the market is overvalued. Based on my calculations, the strategy generates an average return of about 4%, compared to about 1% for a buy-and-hold strategy. Therefore, the CAPE-based market-timing strategy slightly outperforms the benchmark.

60/40 Benchmark Return

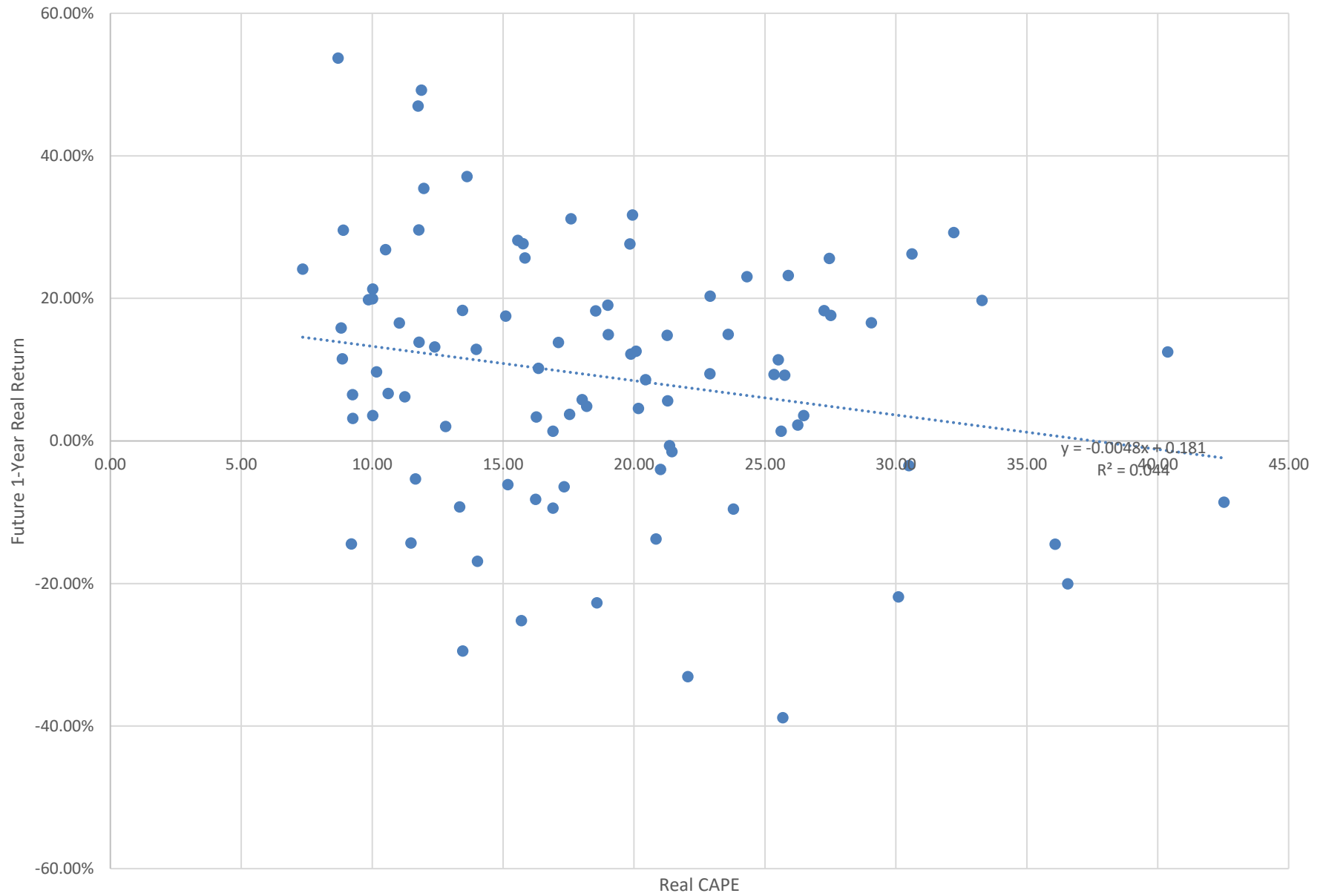
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power over longer horizons. Therefore, I
ation, I allocate 80% to the S&P 500 and
bills. When CAPE is high, indicating
bills.
valued and reduce exposure when it is
out 3% for the static 60/40 benchmark.
enchmark.

CAPE vs 1-Year Return



CAPE vs 5-Year Return

